

Naive Bayes Classifier | Part 5 | Problem Based on Bayes Theorem

1 Problem Setup (Typical Example)

Let's say we want to classify whether an email is **Spam** or **Not Spam** based on the word "Offer".

We have training data:

- 40% of all emails are **Spam** $\rightarrow P(\text{Spam})=0.4$
- 60% are **Not Spam** $\rightarrow P(\text{NotSpam})=0.6$.

Now, from observation:

- Probability that "Offer" appears in Spam = $P(\text{Offer}|\text{Spam})=0.7$
- Probability that "Offer" appears in Not Spam = $P(\text{Offer}|\text{NotSpam})=0.1$.

Question: If an email contains the word *Offer*, what is the probability it is Spam?

2 Applying Bayes' Theorem

We want:

$$P(\text{Spam}|\text{Offer}) = \frac{P(\text{Offer}|\text{Spam}) \cdot P(\text{Spam})}{P(\text{Offer})}$$

Where:

$$P(\text{Offer}) = P(\text{Offer}|\text{Spam}) \cdot P(\text{Spam}) + P(\text{Offer}|\text{NotSpam}) \cdot P(\text{NotSpam})$$

Step 1: Calculate denominator (Evidence)

$$P(\text{Offer}) = (0.7)(0.4) + (0.1)(0.6)$$

$$P(\text{Offer}) = 0.28 + 0.06 = 0.34$$

Step 2: Apply Bayes

$$P(\text{Spam}|\text{Offer}) = \frac{0.7 \cdot 0.4}{0.34}$$

$$P(\text{Spam}|\text{Offer}) = \frac{0.28}{0.34} \approx 0.823$$

So, there is an **82.3% chance** the email is Spam if it contains the word "Offer".

3 Interpretation

- Without the word “Offer”, spam chance was just **40% (prior)**.
- After observing the evidence “Offer”, the chance jumps to **82% (posterior)**.
- This shows how **Bayes’ Theorem updates beliefs with evidence**.

4 Key Points You Should Remember

- Step 1: Write down **priors** ($P(\text{Spam})$, $P(\text{NotSpam})$).
- Step 2: Write **likelihoods** ($P(\text{Word}|\text{Spam})$, $P(\text{Word}|\text{NotSpam})$).
- Step 3: Calculate **evidence** ($P(\text{Word})$).
- Step 4: Apply Bayes formula to get **posterior** ($P(\text{Class}|\text{Word})$).
- Step 5: Choose the class with the **highest probability**.

✓ This is exactly how **Naive Bayes Classifier** works when classifying text, emails, or any other categorical data — except with multiple features instead of just one word.
